FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IV

JEE (Advanced)-2019

PAPER – 2

Time Allotted: 3 Hours **Maximum Marks: 231**

General Instructions:

- The test consists of total 69 questions.
- Each subject (PCM) has 23 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-C & Section-D.**

Section-A (01 – 03, 24 – 26, 47 – 49) *contains 9 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-A (04 – 08, 27 – 31, 50 – 54) *contains 15 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 mark*** *for wrong answer.*
Partial Marks **+1** for each correct option provided no incorrect options is selected.

Section-A (09 – 10, 32 – 33, 55 – 56) *contains 3 paragraphs. Based upon paragraph, 2 multiple choice questions have to be answered. Each question has* ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Section-C (11 – 20, 34 – 43, 57 – 66) *contains 30 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

Section-D (21 – 23, 44 – 46, 67 – 69) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

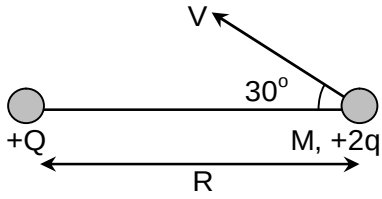
Physics

PART – I

SECTION – A
(Single Correct Choice Type)

This section contains **3 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. In the diagram shown, the charge +Q is fixed. Another charge + 2q, is projected from a distance R from the fixed charge. The minimum separation between the two charges when the velocity becomes $\frac{1}{\sqrt{3}}$ times of the projected velocity, is (Assume gravity to be absent)

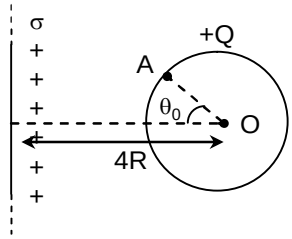


(A) $\frac{\sqrt{3}}{2}R$ (B) $\sqrt{3}\ R$

(C) $\frac{1}{2}R$ (D) 4 R

2. A conducting sphere of radius R and charge Q is placed near a uniformly charged non-conducting infinitely large thin plate having surface charge density σ. Then find the potential at point A (on the surface of sphere) due to charge on sphere

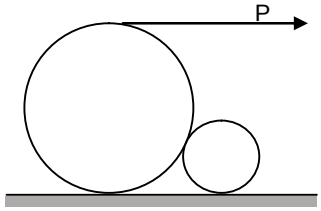
$\left(\text{here } K = \frac{1}{4\pi \in_0}, \theta_0 = \frac{\pi}{3}\right)$



(A) $K\frac{Q}{R} - \frac{\sigma}{4 \in_0}R$ (B) $K\frac{Q}{R} - \frac{\sigma}{2 \in_0}$

(C) $K\frac{Q}{R}$ (D) None of these

3. Two heavy right circular rollers of radii R and r respectively rest on a rough horizontal plane as shown in figure. The larger roller has a string wound around it to which a horizontal force P can be applied as shown. Assuming that the coefficient of friction μ has the same value for all surfaces of contact and the smaller cylinder should neither roll nor slide. The minimum coefficient of friction so that the larger roller can be pulled over the smaller one is



(A) $\frac{r}{R}$ (B) $\left(\frac{r}{R}\right)^2$

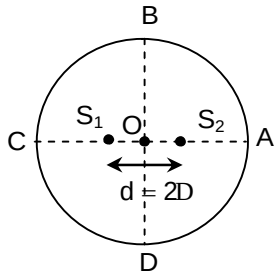
(C) $3\sqrt{\frac{r}{R}}$ (D) $\sqrt{\frac{r}{R}}$

Space for Rough work

(One or More than one correct type)

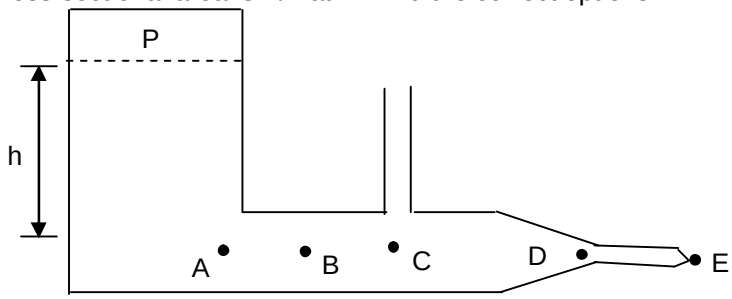
This section contains **5 questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

4. Two point coherent sources separated by d = 2D emits light of wavelength D in phase. A circular wire of radius $R(R >> D)$ has its centre at point O.



(A) Points A, B C and D are points of maxima
(B) Total number of maxima on screen is eight
(C) Total number of minima on screen is eight
(D) None of the above

5. As shown in the figure a liquid of density ρ is standing in a sealed container to a height h. The container contains compressed air at a gauge pressure of P. The horizontal outlet pipe has a cross-sectional area is A/2 at E. Find the correct options.



(A) The velocity of liquid at C will be $\left[\frac{(P+\rho gh)}{4\rho}\right]^{1/2}$

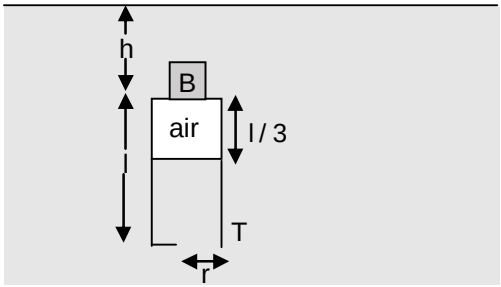
(B) The velocity of liquid at C will be $\left[\frac{2(P+\rho gh)}{\rho}\right]^{1/2}$

(C) The discharge rate is given by $\frac{A}{2\rho}(P+\rho gh)^{1/2}$

(D) The discharge rate is given by $\frac{A}{2\sqrt{\rho}}(P+\rho gh)^{1/2}$

Space for Rough work

6. A light cylindrical tube 'T' of length λ and radius 'r' containing air is inverted in water (density d). One end of the tube is open and the other is closed. A block 'B' of density 2d is kept on the tube as shown in the figure. The tube stays in equilibrium in the position shown. (Assume the atmospheric pressure is to be P_0.). Pick up the correct statement(s).



(A) the volume of block B is $\frac{\pi r^2 l}{3}$

(B) the volume of block B is $\frac{2\pi r^2 l}{3}$

(C) the pressure of air trapped in the tube is $P_0 + dg\left(h + \frac{l}{3}\right)$

(D) the pressure of air trapped in the tube is $P_0 + dg\left(h + \frac{2l}{3}\right)$

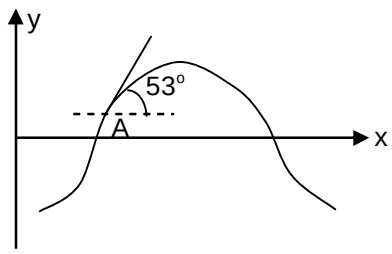
7. A rod of copper, uniform along its length l and of a rectangular cross section of sides of length a and width b(<a) has one end maintained at 100^oC and the other end at 0^oC. The rod is insulated so that no heat is lost from the sides. Let Q_1 be the amount of heat per second that is transmitted along its length, Q_a the heat/s transmitted parallel to a and Q_b the heat/s transmitted parallel to b across any section after the steady state conditions are reached. Then,

(A) $Q_1 = \text{constant}, Q_a > Q_b \neq 0$ (B) $Q_1 = 0, Q_a = Q_b \neq 0$

(C) $Q_1 \neq 0, Q_a = Q_b = 0$ (D) $Q_1 = \text{constant}$

Space for Rough work

8. The y-x curve at an instant for a wave travelling along x axis on a string is shown. Slope at the point A on the curve, as shown, is 53°. Which of the following is/are correct?



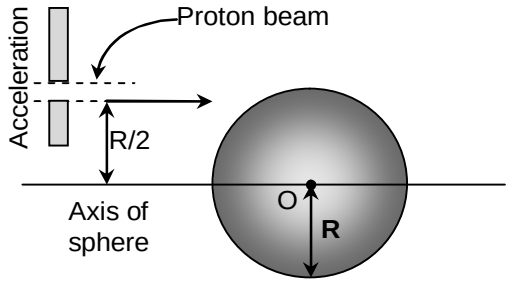
(A) Transverse velocity of the particle at point A is positive if the wave is travelling along positive x axis.
(B) Transverse velocity of the particle at point A is positive if the wave is travelling along negative x axis of the particle at point A.
(C) Magnitude of transverse velocity of the particle at point A is greater than wave speed.
(D) Magnitude of transverse velocity of the particle at point A is lesser than wave speed.

Paragraph type (One Option Correct)

This section contains **1 paragraph** each describing theory, experiment, data etc. **Two questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Questions 9 & 10

An accelerator produces a narrow beam of protons, each having an initial speed of v_o. The beam is directed towards an initially uncharged distant metal sphere of radius R and centered at point O, The initial path of the beam is parallel to the axis of the sphere at a distance of (R/2) from the axis, as indicated in the diagram.



The protons in the beam that collide with the sphere will cause it to become charged. The subsequent potential field at the accelerator due to the sphere can be neglected. The angular momentum of a particle is defined in a similar way to the moment of a force. It is defined as the moment of its linear momentum; linear momentum replacing the force. We may assume the conserved, Assume the mass of the proton as m_p and the charged on it as e. Given that the potential of the sphere increases with time and eventually reaches a constant value.

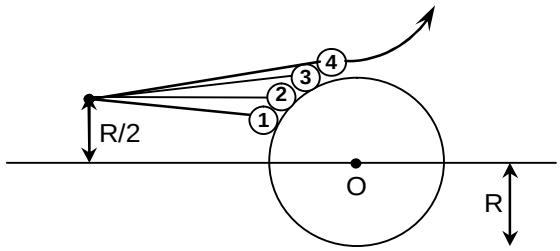
9. The total energy (E) of a proton in the beam travelling with speed v at a distance of $r(r \geq R)$ from point O. Assuming that the sphere has acquired an electrostatic charge Q is

(A) $\dfrac{eQ}{4\pi\varepsilon_0 r}$ (B) less than $\dfrac{eQ}{4\pi\varepsilon_0 r}$

(C) greater than $\dfrac{eQ}{4\pi\varepsilon_0 r}$ (D) zero

Space for Rough work

10. After a long time, when the potential of the sphere reaches a constant value, the trajectory of proton is correctly sketched as
(A) 1
(B) 2
(C) 3
(D) 4

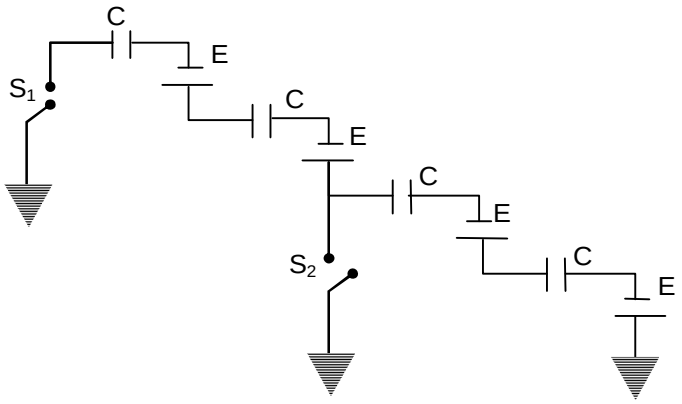


SECTION – C
(Single digit integer type)

This section contains **10 questions**. The answer to each question is a **Single Digit Integer** ranging from 0 to 9, both inclusive.

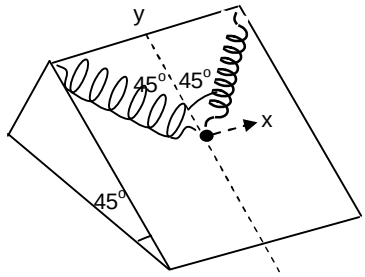
11. Binding energy per nucleon of a fixed nucleus X^A is 6 MeV. It absorbs a neutron moving with KE= 2 MeV, and converts into Y, emitting a photon of energy 1 MeV. If the Binding Energy per nucleon of Y (in MeV) is found to be $\frac{(xA-1)}{(A+1)}$. Find the value of x.

12. In the given circuit, all the capacitors are initially uncharged. After closing the switch S_1 for a long time suddenly S_2 is also closed and kept closed for a long time. Find the total heat produced after closing S_2 (in joules).

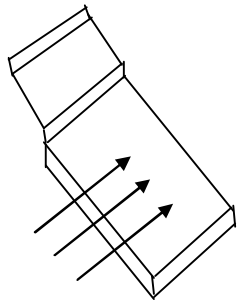


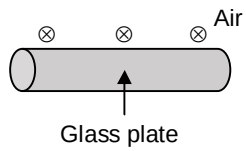
Space for Rough work

13. Figure shows a ball of mass m connected with two ideal springs of force constant k, kept in equilibrium on a smooth incline, suddenly right spring is cut. What is magnitude of instantaneous acceleration (in m/s^2) of ball?

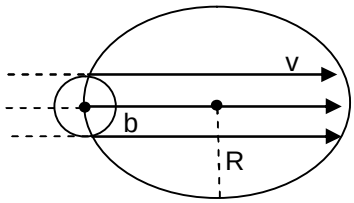


14. A thin glass plate of area A and mass m is hinged along one of its sides. The speed with which the air should be blown parallel to its upper surface to hold the plate horizontal is $\gamma\sqrt{\frac{mg}{18pA}}$. Find the value of γ. The density of air is p.

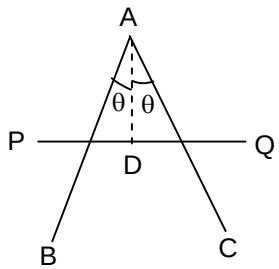




15. A soap film is made by dropping a circular frame of radius b in soap solution. A bubble is formed by blowing air with speed v in the form of a cylinder. Air stops after striking surface of soap bubble. Density of air is ρ. The radius of the bubble is R > b. The radius R of the bubble when the soap bubble separates from the ring is $\frac{\beta T}{\rho v^2}$. Find the value of β (where surface tension of liquid is T).

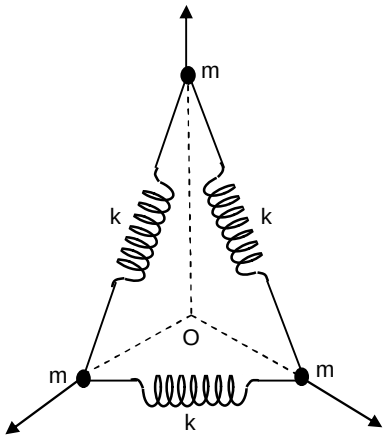


16. In the figure shown AB, BC and PQ are thin, smooth, rigid wires. AB and AC are joined at A and fixed in vertical plane. $\angle BAC = 2\theta = 9^o$ and line AD is angle bisector of angle BAC. A liquid of surface tension $T = 0.205\,N/m$ forms a thin film in the triangle formed by intersection of the vires AB, AC and PQ. In the figure shown the uniform wire PQ of mass 1 gm is horizontal and in equilibrium under the action of surface tension and gravitational force. Find the time period of SHM of PQ in vertical plane for small displacement from its mean position, in the form $\frac{\pi}{X}s$ and fill value of X.



Space for Rough work

17. Three identical ideal springs, each of force constant k are joined to three identical balls (each of mass m) as shown in the figure. O is centroid of the triangle. Initially, each of the springs is in its natural length. Now all the three balls are simultaneously given small displacements of equal magnitude along the directions shown in the figure. The oscillation frequency of balls is $\frac{1}{2\pi}\sqrt{\frac{\beta k}{2m}}$. Find the value of β.



18. An ant with mass m is standing peacefully on top of a horizontal, stretched rope. The rope has mass per unit length μ and is under tension F. Without warning, a student starts a sinusoidal transverse wave of wavelength λ propagating along the rope. What minimum wave amplitude (in mm) will make the ant feel weightless momentarily? Assume that m is very small and that the presence of the ant has no effect on the propagation of the wave. (Given: $\lambda = 0.5\text{m}, \mu = 0.1\,\text{kgm}^{-1}, F = 3.125\text{N}, g = \pi^2$).

19. In a car race sound signals emitted by the two cars are detected by the detector on the straight track at the end point of the race. Frequency observed are 330 Hz and 360 Hz and the original frequency is 300 Hz of both cars. Race ends with the separation of 100 m between the cars. Assume both cars move with constant velocity and velocity of sound is 330 m/s. Find the time (in sec) taken by winning car.

20. Two identical wires are stretched by the same tension of 100 N and each emits a note of frequency 200 Hz. The tension in one wire is increased by 1 N. Calculate the number of beats heard per second when the wires are plucked.

Space for Rough work

SECTION – D
(Numerical Answer Type)

This section contains **3 questions**. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. xxxxx.xx).

21. A glass beaker of mass 400 g floats in water with the open end just touching the surface of water and half of the beaker filled with water. The inner volume of the beakers is 500 cm^3. What is the density (in g cm^{-3}) of the material of beaker?

22. The equation for the vibration of a string fixed at both ends vibrating in its third harmonic is given by y = 2 cm sin $\left[\left(0.6\ cm^{-1}\right)x\right]\cos\left[\left(500\pi\ s^{-1}\right)t\right]$ The length of the string is:

23. An open organ pipe containing air resonates in fundamental mode due to a tuning fork. The measured values of length $l\ (\text{in cm})$ of the pipe and radius r (in cm) of the pipe are $l = 94 \pm 0.1$ r = 5 ± 0.05. The velocity of the sound in air is accurately known. The maximum percentage error in the measurement of the frequency of that tuning fork by this experiment, will be

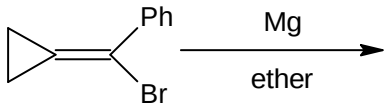
Space for Rough work

Chemistry

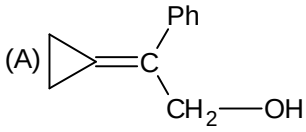
PART – II

SECTION – A
(Single Correct Choice Type)

This section contains **3 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

24. 

B; Prodcut B is

(A) 

(B) $Ph—C\equiv C—CH_2—CH_2—CH_2OH$

(C) $Ph—C\equiv C—\underset{\displaystyle OH}{\underset{|}{CH}}—CH_3$

(D) $Ph—CH_2C\equiv C—CH_2—CH_2—OH$

25. In a unit cell of AB if all the atoms along one of the tetrad axis of symmetry and one of the diad axis of symmetry are removed, then the resultant formulae of the unit cell will be (Atoms 'A' occupy all T.V. and atoms 'B' form fcc)
(A) A_5B_6
(B) A_5B_{11}
(C) A_2B_3
(D) No change in formula.

26. Consider the following reactions
(I) $Na_2S_2O_3 + \text{dil HCl} \xrightarrow{\Delta}$
(II) $Na_2S_2O_3 + Cl_2 + H_2O \xrightarrow{\Delta}$
(III) $SO_2 + H_2S \xrightarrow{\Delta}$
(IV) $H_2S + H_2O_2 \xrightarrow{\Delta}$
The reactions which give yellow turbidity are
(A) II, III & IV
(B) III & IV
(C) I, III & IV
(D) I, II, III & IV

Space for Rough work

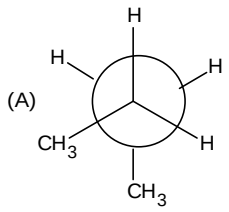
(One or More than one correct type)

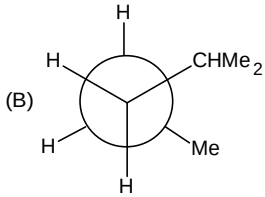
This section contains **5 questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

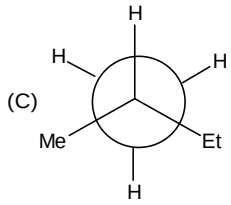
27. For aq. solution of potash alum, which is/true are
(A) One gm eq of potash alum will give 1 gm eq of K^+
(B) One gm eq of potash alum will give 1 gm eq of Al^{3+}
(C) One gm eq of potash alum will give 2 gm eq of SO_4^{2-}
(D) $\frac{1}{8}$ mole of potash alum will give 1 gm eq of SO_4^{2-}

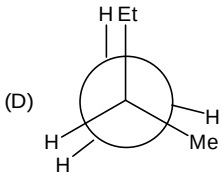
28. If a coordination complex $M(CO)_6$ is colored due to d – d transition & its CFSE corresponds to that wavelength which an electron emits on de – excitation from 4th orbit to 3rd orbit in Li^{2+} ion. Find the value of CFSE for the given complex. $\left(\text{take } R_\infty = 10^7 m^{-1}\right) Ch = 6.62 \times 10^{-34}$
(A) 9.53×10^{-19} J (B) 1.36×10^{-18} J
(C) 1.97×10^{-20} J (D) 2.59×10^{-17} J

29. Which of the following Newmann projection is correct for 2 – methyl butane?

(A)


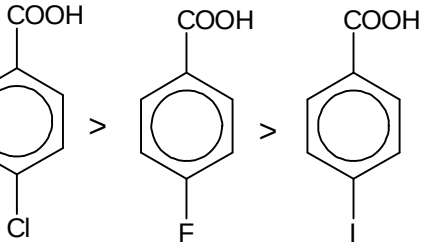
(B)


(C)


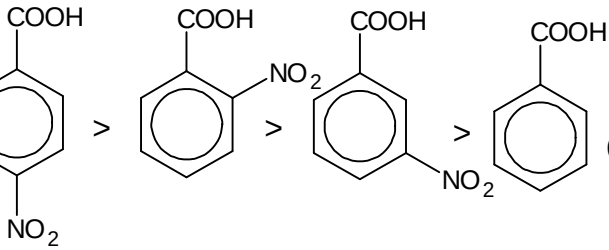
(D)


Space for Rough work

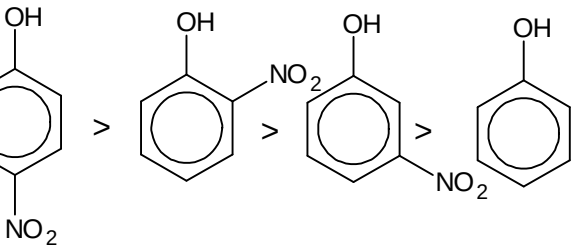
30. Which of the following is correct order as per the mention property?

(A) 

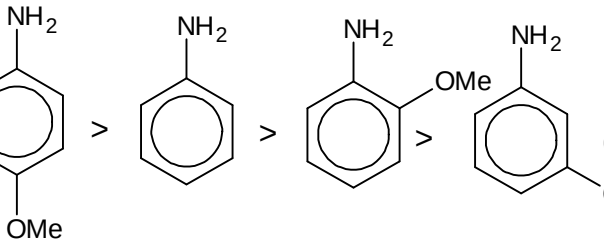
(acidic strength)

(B) 

(acidic strength)

(C) 

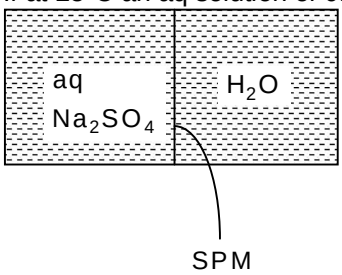
(acidic strength)

(D) 

(basic strength)

Space for Rough work

31. If at 25^oC an aq solution of 0.4 M Na_2SO_4 and water are separated by SPM as mentioned below



If a pressure of 2 atm is applied on solution, find what will happen.
(A) Reverse osmosis takes place
(B) Osmosis takes place
(C) Na_2SO_4 will move from solution to solvent side
(D) Nothing will happen

Paragraph type (One Option Correct)

This section contains **1 paragraph** each describing theory, experiment, data etc. **Two questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Questions 32 & 33

$$\underset{(\text{alkali Metal})}{M} + N_2 \longrightarrow A$$

$$A + H_2O \longrightarrow B + C\uparrow$$

$$C + CaOCl_2 \longrightarrow D\uparrow + E + F$$

$$D + CaC_2 \xrightarrow{1100^oC} C + H$$

$$H + O_2 \xrightarrow{\text{burn}} \text{One compound } (\text{gas}) \text{ is proudced}$$

$$C + H_2O \longrightarrow I + \underset{\text{lower mol.wt as compare to C}}{J}$$

32. Which is true for 'J'?
(A) H is metal oxide
(B) H gives Nesseler's test
(C) Salt of H with strong base undergoes anionic hydrolysis to give alkaline solution
(D) H can be obtained by treating D with H_2O

Space for Rough work

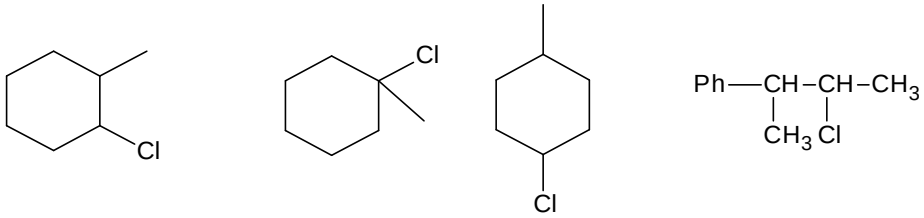
33. Which of the following statements is correct?
(A) M is Na Metal
(B) C can be obtained by strong heating of $(NH_4)_2Cr_2O_7$
(C) The gas produced by combustion of H is also produced by heating compound 'I'
(D) Gas D can not be obtained by heating NH_4NO_3

SECTION – C
(Single digit integer type)

This section contains **10 questions**. The answer to each question is a **Single Digit Integer** ranging from 0 to 9, both inclusive.

34. Compound (A) gives positive Lucas test in 5 min. when 6.0 gm of (A) is reacted with Na metal, 1120 mL of H_2 is evolved at STP. Assuming A to contain one oxygen per molecule, the number of carbon atom per molecule in (A) will be_____.

35. How many of following will give same SN^1 & SN^2 products (*excluding stereo isomers*) ?



36. Calculate the pH at which an acid indicator with $ka = 1\times10^{-5}$ changes color when indicator concentration is $1\times10^{-3}\ M$.

37. 100 mL of hydrocarbon is mixed with excess of oxygen and exploded. Our cooling the mixture was reported to have contraction of 250 mL the remaining gas when passed through a solution of eq. KOH, The mixture shows a further contraction of 300 mL. Find the number of carbon atoms per molecule of hydrocarbon.

38. The critical temperature and pressure CO_2 gas are 304.2 k and 72.9 atm respectively. If the radius of CO_2 molecules is 1.62×10^{-x}cm. Assuming it to behave as Vander Waal's gas find x.

39. 30 mL of a solution containing 9.15 gm/lt of an oxalate $K_xH_y(C_2O_4)_z.nH_2O$ are required for titrating 27 mL of 0.12 N NaOH and 36 ml of 0.12 M $KMnO_4$ separately x + y + z is_____.

Space for Rough work

40. A 2.18 gm sample contain a mixture of XO and X_2O_3. If reacts with 0.015 moles of $K_2Cr_2O_7$ to oxidize the sample completely to form XO_4^- and Cr^{+3}. If 0.0187 moles of XO_4^- is formed; the atomic mass of x was found to by approx. $a \times 10^2$. Calculate a.

41. If Ma_3b_3 has x pairs of enantiomers,
$M(AA)_3$ has y pairs of enantiomers &
$Ma_2b_2c_2$ has z pairs of enantiomers
Find the value of x + y + z.
(a,b & c are monodentate & AA is symmetrical bidentate ligand)

42. In how many of the following reactions black ppt is produced?

(i)	$Ag + O_2 + H_2S \longrightarrow$	(v)	$\underset{aq}{CuSO_4} + H_2S \longrightarrow$
(ii)	$Ag + H_2S \longrightarrow$	(vi)	$\underset{aq}{CdSO_4} + H_2S \longrightarrow$
(iii)	$\underset{aq}{Hg^{2+}} + H_2S \longrightarrow$	(vii)	$AgNO_3 + \text{dil. } HCl \longrightarrow$
(iv)	$\underset{aq}{PbCl_2} + H_2S \longrightarrow$	(viii)	$Fe^{3+} + NH_4Cl + NH_4OH \longrightarrow$

43. Which of the following species can be oxidized by O_3?
$HF, HCl, PbS, As(\text{in water}), I_2, K_2MnO_4, Hg, SO_2, KI, K_2SO_4$

Space for Rough work

SECTION – D
(Numerical Answer Type)

This section contains **3 questions**. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. xxxxx.xx).

44. If a sample of 2 – bromopentane shows specific rotation of + 30° & specific rotation of optically pure 2 – bromobentane in the 40°, find % of (d) 2 – bromopentane in the given sample.

45. If Cu atoms form fcc (ABC ABC……..) and nearest distance between two 'A' layer is $8\sqrt{6}$ nm then find the radius of atoms (in nm) that exactly fits into the octahedral void (do not round off)

46. How many of the following reactions will release a gas as any product (at 25^oC)?

(i)	$NH_4^+ + \overline{O}H \longrightarrow$	(vi)	$PH_3 + NO_2 \xrightarrow{\text{electric spark}}$
(ii)	$Al + NaOH + H_2O \longrightarrow$	(vii)	$NH_3 + CaOCl_2 \longrightarrow$
(iii)	$Al(OH)_3 + NaOH \longrightarrow$	(viii)	$SOCl_2 + H_2O \longrightarrow$
(iv)	$Ba + \text{cold water} \longrightarrow$	(ix)	$SnCl_2 + HCl + O_3 \longrightarrow$
(v)	$(NH_4)_2 Cr_2O_7 \xrightarrow{\Delta}$		

Space for Rough work

Mathematics

PART – III

SECTION – A
(Single Correct Choice Type)

This section contains **3 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

47. Let $a_1 = 1$; and $a_n = n(a_{n-1} + 1)$ for n = 2, 3,..... define $P_n = \left(1+\frac{1}{a_1}\right)\left(1+\frac{1}{a_2}\right).....\left(1+\frac{1}{a_n}\right)$

Then $\lim_{n\to\infty} P_n$ is

(A) 1 + e (B) e
(C) 1 (D) ∞

48. If the roots of the equation $x^4 + ax^3 + bx^2 + cx + d = 0$ are in geometric progression, then

(A) $b^2 = ac$ (B) $a^2 = b$
(C) $a^2b^2 = c^2$ (D) $c^2 = a^2d$

49. $f'(x) = \frac{3^{1/x}}{1+3^{1/x}}$ then $\int_{-1}^{1} f'(x)dx = \alpha$ and $\int_{-1}^{1} xf''(x)dx = \beta$

(A) $\alpha = 1,\ \beta = 1/2$ (B) $\alpha = 1/2,\ \beta = 0$
(C) $\alpha = -\frac{1}{2},\ \beta = 1$ (D) $\alpha = 1/2,\ \beta = -1$

(One or More than one correct type)

This section contains **5 questions**. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

50. If a tangent on ellipse at A (1, 1) intersect its directrix at B (7, –6) and S be the focus of ellipse and C (α, β) is the centre of ΔSAB, then

(A) $\alpha + \beta = 1$ (B) $\alpha - \beta = 7$
(C) $SC^2 = 20.5$ (D) $SC^2 = 21.25$

Space for Rough work

51. If curve $A \equiv \{z : |z-3|+|z+3| = 8\}$ & $B \equiv \{z : |z-3| = K\}$, $K \in R^+$ touches the curve 'A' internally, and given another curve $C \equiv \{z : \|z-3|-|z+3\| = 4\}$, then;
(A) A & C intersect orthogonally & k = 1
(B) K = 1
(C) K = 3/2
(D) B touches also C internally

52. If $f^3(x) - 3x^2 f(x) + x^3 = 0$, then $\int \frac{f(x)}{x} dx$ is equal to
(A) $f(x) + c$ (B) $-f(x) + c$
(C) $2f(x) + c$ (D) $xf'(x) + c$

53. If $A = \begin{pmatrix} \alpha & 0 \\ 2 & \alpha \end{pmatrix}$ and $\det(2A^2 - 2A) = 144$, then possible correct statement is (for real part)
(A) $\text{tr}(A) = 6$ (B) $\det(A) = 9$
(C) $\det(A) = 4$ (D) $\text{tr}(A) = 1$

54. If $\vec{u}(x) = x\hat{i} + (\sin x)\hat{j} + (1+\sin x)\hat{k}$, $\vec{v}(x) = \hat{i} + \cos x(\hat{j} + \hat{k})$ and $\vec{w}(x) = \sin x(\hat{j} + \hat{k})$, all are linearly dependent vectors for $x = \alpha$, then $\left(\frac{2\alpha}{\pi}\right)$ can be equal to
(A) 101 (B) 1001
(C) 2012 (D) 2011

Space for Rough work

Paragraph type (One Option Correct)

This section contains **1 paragraph** each describing theory, experiment, data etc. **Two questions** relate to the paragraph. Each question of a paragraph has **Only One correct** answer among the four choices (A), (B), (C) and (D).

Paragraph for Questions 55 & 56

If f(x) & g (x) are two real valued functions such that

$f(\sin x)+f(\cos x)=2x-\frac{\pi}{2}$ and $F(x)=\begin{cases}\frac{8}{3f'(x)} & \text{for } x^2\neq 1\\ 0 & \text{for } x^2=1\end{cases}$ & $g(x)=\frac{3}{4}F(x)$.

55. The range of $f(x)$ is

(A) $\left[-\pi,\ \frac{3\pi}{2}\right]$ (B) $\left[-\frac{3\pi}{2},\ \pi\right]$

(C) $\left[-\frac{3\pi}{2},\ \frac{\pi}{2}\right]$ (D) none of these

56. The only one solution of $f(x)=\sin\left(\frac{x}{2}+\frac{\pi}{4}\right)$ for $-\frac{3\pi}{2}\le x\le\frac{\pi}{2}$ is β. Then

(A) $1>\beta>\alpha$ (B) $-1<\beta<\alpha$

(C) $\beta>1$ (D) $\beta=1$

SECTION – C
(Single digit integer type)

This section contains **10 questions**. The answer to each question is a **Single Digit Integer** ranging from 0 to 9, both inclusive.

57. If $x^2+y^2=16$ is director circle of locus of centre of a moving ellipse touches both the axes in first quadrant, then if major axis and minor axis are not fixed, then maximum value of product of semi-major and semi-minor axis is

Space for Rough work

58. If $f(x)=\sum_{i=1}^{\infty}\frac{x}{\{(i-1)x+1\}(ix+1)}$, then $f(2017)+f\left(\frac{1}{2017}\right)$ is equal to

59. If $\cos^{-1}(2-x^2)+\sin^{-1}(2-x^2)+\tan^{-1}x\geq\frac{2\pi}{3}$ have no solution if $x<\alpha$ or $x>\beta$ $(\text{for } \alpha<\beta)$, then $\left\{\frac{1}{\max(\alpha)}+(\min(\beta))^2\right\}=$ .....

60. If $I=\int_1^2\frac{x^3-3x^2}{\ln(x-1)}dx$ and $J=\int_2^3\frac{(3x-5)}{\ln(x-2)}dx$, then e^{I+J} equals .....

61. Position vector of a point A is $\left(a\hat{i}+b\hat{j}+c\hat{k}\right)$ and $a^2+b^2+c^2=4$; and position vector of another point B is $\left(p\hat{i}+q\hat{j}+r\hat{k}\right)$ and $p^2+q^2+r^2=9$. Given $\angle AOB=2\cos^{-1}\frac{1}{5}$, where O is origin. If AQ is perpendicular to angle bisector of $\angle AOB$ i.e. line OQD where D lies on AB, then value of $25|QD|$ = ?

62. If $\frac{3x^2+1}{\sqrt{x^4+x^2}}\geq f(x)$, where $f(x)$ is a constant function in $0<x\leq\sqrt{2}$, then minimum value of $\int_0^{\sqrt{2}}\frac{(3x^2+1)dx}{\sqrt{x^4+x^2}}$ is .....

63. If $AB=BA=I$ such that $A=\left[a_{ij}\right]_{2\times2}$, $B=\left[b_{ij}\right]_{2\times2}$, $a_{ii}=b_{ii}$, $a_{12}=-b_{12}$, $a_{21}=-b_{21}$ and a_{ij} are co prime natural numbers, then $|a_{21}-a_{12}|=$ .....

Space for Rough work

64. If $f'(x) > 0 \ \forall \ x \in R^+$ where $f : R^+ \to R$ and $f(x) + \frac{1}{x} = f^{-1}\left(\frac{1}{f(x)}\right)$ and $f^{-1}\left(\frac{1}{f(x)}\right) > 0$ then $\frac{\pi}{\sin^{-1}(f(2))}$ is equal to

65. If S_r is sum of infinite G.P. $(r \geq 3)$, where first term is $\frac{(r-2)}{(r-1)!}$ and common ratio $\frac{1}{(r-1)}$, then $1 + \frac{2011}{(2009)!} + \sum_{r=3}^{r=2011} \left|(r^2 - 4r + 2) \times S_r\right|$ is equal to .....

66. If $\lim_{x \to 0} \frac{a\tan x - b\sin x + cx^2 - x}{x^3} = \frac{3}{2}$ then, $[a + b + c]$ is equal to _____ . (where [.] denotes greatest integer part)

SECTION – D
(Numerical Answer Type)

This section contains **3 questions**. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded - off to the **second decimal place**; e.g. xxxxx.xx).

67. The length of focal chord AB of ellipse $\frac{x^2}{4} + \frac{y^2}{3} = 1$ is: $\left(\text{Given } A = \left(\frac{8}{5}, \frac{3\sqrt{3}}{5}\right)\right)$

68. How many matrices X with entries {0, 1, 2} are there for which sum of diagonal entries of $X.X^T$ is 7?

69. If $\log_{(3x+4)}(4x^2 + x + 1) + \log_{(2x+1)}(6x^2 + 11x + 4)$ then x is _____.

Space for Rough work